CCEE Preparation Cheatsheet

Cloud Computing & Engineering Exam Guide

1. Amazon Web Services (AWS) Core Concepts

Elastic Compute Cloud (EC2)

EC2 allows you to rent virtual computers (instances) to run applications. It is the primary **IaaS** (Infrastructure as a Service) offering.

- **AMI (Amazon Machine Image):** The "template" for your VM. Contains the OS (Linux, Windows) and pre-installed software.
- **Instance Type:** Determines the hardware capacity (CPU, RAM, Storage type).
- **Security Groups:** Acts as a **virtual firewall** at the instance level. You must explicitly allow traffic (e.g., allow Port 22 for SSH, Port 80 for Web).
- **Access Methods:**
  - **Linux:** Accessed via **SSH** (Secure Shell) using a private key pair (`.pem` file).
  - **Windows:** Accessed via **RDP** (Remote Desktop Protocol).

Virtual Private Cloud (VPC)

A logically isolated section of the AWS cloud where you launch resources in a virtual network.

- **Subnets:** Segments of a VPC's IP address range.
  - *Public Subnet:* Has direct access to the internet.
  - *Private Subnet:* No direct internet access; used for backend databases.
- **Internet Gateway (IGW):** The component that allows communication between your VPC and the internet.

Storage & Serverless

- **AWS Lambda:** Serverless compute. Run code without provisioning or managing servers. You pay only for the compute time consumed.
- **S3 (Simple Storage Service):** Object storage for infinite scale. Used for backups, static website hosting, and media files.

2. Microsoft Azure Core Concepts

Management Tools

- **Azure Portal:** Web-based GUI for managing all Azure services.
- **Azure CLI / PowerShell:** Command-line tools for scripting and automation.

Azure Storage Types

- **Blob Storage:** Object storage for unstructured data (images, videos, logs). Direct equivalent to AWS S3.
- **File Storage:** Managed file shares accessible via the SMB protocol (acts like a network drive).
- **Azure SQL Database:** Fully managed relational database service (PaaS).
- **Cosmos DB:** Globally distributed, multi-model NoSQL database.

Azure Functions

Serverless compute service (FaaS). Allows you to run event-triggered code without infrastructure management. Equivalent to AWS Lambda.

3. Quick Comparison: AWS vs. Azure

Feature/Category	Amazon Web Services (AWS)	Microsoft Azure
Virtual Machines	Amazon EC2	Azure Virtual Machines
Serverless Compute	AWS Lambda	Azure Functions
Object Storage	Amazon S3	Azure Blob Storage
Virtual Network	VPC (Virtual Private Cloud)	VNet (Virtual Network)
NoSQL Database	Amazon DynamoDB	Azure Cosmos DB
SQL Database	Amazon RDS	Azure SQL Database
Identity Mgmt	AWS IAM	Azure Active Directory (Entra ID)

4. Critical Study Concepts

- **Shared Responsibility Model:**
  - *Cloud Provider:* Responsible for "Security **of** the Cloud" (Hardware, global infrastructure, networking components).
  - *Customer:* Responsible for "Security **in** the Cloud" (Customer data, OS patches, firewall configuration, IAM).
- **Hands-on Workflow (VM Creation):** *Select Image (OS)* $\rightarrow$ *Choose Size (CPU/RAM)* $\rightarrow$ *Configure Network (VPC/VNet)* $\rightarrow$ *Set Security Rules (Ports)* $\rightarrow$ *Connect (SSH/RDP).*